A preparation dictionary for the selected state

Modular spectra and energy-resolved balance

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Abstract

The complete selected bilocal mean and covariance call for a distinction between preparation and the generator of static-patch time. We apply standard tracial modular theory to the already established selected density. Its modular flow has an explicit q-Mehler spectral representation after an orthogonal rotation of the two q-Gaussian fields. We derive a nonzero stationary part in the modular correlation of the microscopic Hamiltonian, spectral sum rules, and a finite-window test that rules out every constant Hamiltonian temperature even on the stationary charged restriction. These are model-level preparation data for a subsequent bulk dictionary; no spatial generator is supplied by taking a logarithm of the density.

1 The physical comparison and its conventions

Miyashita–Sekino–Susskind identify the microscopic Hamiltonian with static-patch boost time, while distinguishing its Boltzmann temperature from the Gibbons–Hawking temperature [1, Sec. 2, (2.4)–(2.5)]. Their flat-space construction uses the maximally mixed microscopic background, whereas the bulk Rindler vacuum has dimensionless temperature $1/(2\pi)$ [2, Secs. 1, 6]. The singlet mass comparison is still a quantitative target [1, Sec. 4.3]. These statements do not assign a new temperature to every prepared microscopic state.

Our model is the same-disorder overlapping complex Gaussian p-body CAR system, at half filling. Exact finite-N pinching precedes $p^2/N \to \lambda > 0$ double scaling. Use

$$q = e^{-4\lambda}, \quad \zeta = \sqrt{q}, \quad C = J_s^2/\lambda, \quad h = H/\sqrt{C}, \quad W = \sqrt{\lambda}U, \quad \hbar = 1.$$

The fields h, U are orthogonal unit-variance q-Gaussians in the faithful tracial limiting algebra. Let $\mathcal{A}$ be the controlled covariant algebra containing them and the charged curvature X. The preceding selected-state and higher-descendant results give

$$\sigma(A) = \tau(\rho A), \qquad \rho = 1 - \zeta(1+\lambda) + \zeta(h-W)^2. \tag{1}$$

All algebraic results below concern this limiting algebra. We do not infer microscopic modular convergence from polynomial moment convergence alone.

Physical Hamiltonian time is $\alpha_T(A) = e^{iTh} A e^{-iTh}$, with $T = \sqrt{C}\, t_{\mathrm{phys}}$. The previous string variable is $u = 2\sqrt{\lambda}T$; the earlier candidate Rindler matching used $\eta = 2u$. The modular parameter introduced below is a separate dimensionless variable. Lin–Stanford's bra/ket maps [3, (158)–(159)] are kept as preparation-dependent maps; they are not a uniform rescaling of all these clocks.

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2 A computable modular flow

Use the already established rotation

$$v = \frac{h - \sqrt{\lambda}U}{\sqrt{1+\lambda}}, \qquad z = \frac{\sqrt{\lambda}h + U}{\sqrt{1+\lambda}}, \qquad h = \frac{v + \sqrt{\lambda}z}{\sqrt{1+\lambda}}. \tag{2}$$

Orthogonal second quantization preserves the q-Gaussian trace [4, Theorem 2.11]; hence v, z have the same joint law as h, U. Set

$$a = 1 - \zeta(1+\lambda), \quad b = \zeta(1+\lambda), \quad R = \frac{2}{\sqrt{1-q}}, \quad p(x) = a + bx^2. \tag{3}$$

Then $\rho = p(v)$, $a + b = 1$ and $a > 0$, since $e^{2\lambda} > 1 + \lambda$. The known exact spectral bounds are $a \le \rho \le a + bR^2$. In particular the state is faithful.

On the tracial standard Hilbert space $L^2(\mathcal{A}, \tau)$, the normalized cyclic separating vector is $\Omega_\sigma = \rho^{1/2}$. For left and right multiplication L, R, standard modular theory [5, Sec. 4.2] gives

$$\Delta_\sigma = L_\rho R_{\rho^{-1}}, \qquad \Delta_\sigma^{is} A \Delta_\sigma^{-is} = \rho^{is} A \rho^{-is}.$$

We choose the opposite sign to use ordinary positive-temperature Heisenberg conventions:

$$k(x) = -\log p(x), \quad K = k(v), \qquad \beta_s(A) = e^{isK} A e^{-isK} = \rho^{-is} A \rho^{is}. \tag{4}$$

Then σ is a $\beta = 1$ KMS state for β_s. The implementing generator on L^2 is $L_K - R_K$, not L_K alone. This follows directly by moving ρ around the trace; no finite-dimensional matrix trace is assumed.

The state also has the exact preparation forms

$$\sigma(A) = a\tau(A) + b\tau(vAv) = \langle \rho^{1/2}, A\rho^{1/2} \rangle_2. \tag{5}$$

These are two descriptions of the same established density. The rotation, positivity and nonstationarity were already proved in [6]; the new calculations below are its modular response and an energy-resolved balance test.

In particular the inherited identity

$$\|[h, \rho]\|_2^2 = 2q\lambda(1+\lambda)(1+q)(1-q^2) > 0$$

shows that no nonzero constant rescaling of α_T equals (4) on the full algebra: β_s preserves σ, whereas α_T does not. The modular clock is available for computing preparation response, not as a replacement boost.

3 Exact modular spectral kernel

Let $d\mu_q$ be the unit-variance q-Hermite measure and let $K_r(x,y)$ denote its Mehler kernel with eigenvalues r^n. The one-z bimodule measure relative to v is

$$d\nu(x,y) = K_q(x,y)d\mu_q(x)d\mu_q(y).$$

For $\mathcal{W}(s) = \sigma(\beta_s(z)z)$, functional calculus and the two-z trace formula give

$$\boxed{\mathcal{W}(s) = \int p(x)e^{is[k(x)-k(y)]}\, d\nu(x,y).} \tag{6}$$

Here x is the state-weighted energy of v, not the original Hamiltonian energy h. The positive spectral measure, using $\mathcal{W}(s) = \int e^{-i\omega s}dS(\omega)$, is the pushforward of $p(x)d\nu$ under

$$\omega = k(y) - k(x) = \log\frac{p(x)}{p(y)}.$$

It is supported in $[-\Omega,\Omega]$, with

$$\Omega = \log\frac{a+bR^2}{a}. \tag{7}$$

Exchanging x, y proves the measure identity $dS(-\omega) = e^{-\omega}dS(\omega)$, equivalently

$$\mathcal{W}(s-i) = \mathcal{W}(-s) \qquad (s\in\mathbb{R}). \tag{8}$$

At fixed q, the kernel is entire in s because K is bounded. This KMS statement fixes the sign convention in (4).

Some useful exact sum rules are

$$\mathcal{W}(0) = 1, \tag{9}$$

$$i\mathcal{W}'(0) = \frac{1}{2}\int [p(x)-p(y)]\log\frac{p(x)}{p(y)}d\nu > 0, \tag{10}$$

$$-\mathcal{W}''(0) = \int p(x)\left[\log\frac{p(x)}{p(y)}\right]^2 d\nu, \tag{11}$$

$$D(\sigma\|\tau) = \int p(x)\log p(x)d\mu_q(x). \tag{12}$$

The entropy is relative to the normalized trace, not an absolute von Neumann entropy of a continuum wedge. Strict positivity in (10) uses the strictly positive kernel and nonconstant p.

The microscopic Hamiltonian has the modular correlation

$$\sigma(\beta_s(h)h) = \frac{1+b(1+q)}{1+\lambda} + \frac{\lambda}{1+\lambda}\mathcal{W}(s). \tag{13}$$

Cross terms vanish by odd z parity, and $\sigma(v^2) = 1+b(1+q)$. The $\mathcal{W}$ spectrum has no atoms: on each half of $[-R,R]$, k is strictly monotone with nonzero derivative away from zero. Its pushforward of the integrable two-variable density is absolutely continuous; the isolated critical point carries no mass. Taking the difference of the two transformed variables preserves absolute continuity. Riemann–Lebesgue therefore gives $\mathcal{W}(s)\to 0$. Thus (13) has the explicit nonzero plateau shown there. This is a modular-time result after fixed-λ double scaling.

4 A finite-window test of Hamiltonian temperature

The previously proved charged restriction on $W^*(h, X)$ is

$$\sigma|_{h,X} = \tau_f, \qquad f(h) = 1 - \zeta + \zeta h^2,$$

where X is the normalized centered charged curvature with Hamiltonian crossing q^2. This restriction is stationary under α_T even though the full state is not. Its unfiltered X autocorrelation is real and even. Consequently that one autocorrelation alone has symmetric frequency weight and cannot establish tracial equilibrium on the whole algebra.

Take a bounded energy-filtered probe $A_{IJ} = P_I(h)XP_J(h)$. The exact same-disorder two-energy measure is $d\nu_X = K_{q^2}(x, y)d\mu_q(x)d\mu_q(y)$, and

$$F_{IJ} := \tau_f(A_{IJ}^\dagger A_{IJ}) = \int_{I\times J} f(y)d\nu_X, \tag{14}$$

$$R_{IJ} := \tau_f(A_{IJ}A_{IJ}^\dagger) = \int_{I\times J} f(x)d\nu_X. \tag{15}$$

These are positive probe norms (or transition weights), not rates obtained from a new interaction model.

Proposition 1. *The restriction τ_f on $W^*(h, X)$ is not a KMS state for α_T at any constant real inverse temperature β.*

Proof. The bounded inner dynamics makes every bounded probe entire analytic. Its KMS identity requires energy-resolved reverse to forward ratio $e^{-\beta(x-y)}$. Equivalently,

$$R_{IJ} = \int_{I\times J} e^{-\beta(x-y)} f(y)d\nu_X.$$

Choose $I_+ = [.8, 1.2]$, $I_- = [-1.2, -.8]$, $J = [-.2, .2]$. All intervals lie inside the support for every $0 < q < 1$. The positive Mehler kernel and monotonicity of f in $|x|$ give

$$\frac{R_{I_+J}}{F_{I_+J}} = \frac{R_{I_-J}}{F_{I_-J}} \geq \frac{1 - \zeta + .64\zeta}{1 - \zeta + .04\zeta} > 1. \tag{16}$$

The equality follows from simultaneous reflection. In the positive window $x - y > 0$, so this inequality would force $\beta < 0$. In the negative window it would force $\beta > 0$. The value $\beta = 0$ would give ratio one in both. No constant temperature is consistent with the two probes. □

At the density level the same obstruction is $f(x)/f(y) \neq e^{-\beta(x-y)}$ for any fixed β on the full support: evenness would force $\beta = 0$, then f would have to be constant. The restricted state's own modular Hamiltonian is instead $K_f = -\log f(h)$. This distinguishes physical stationarity from thermal balance, and uses actual probes in the charged algebra rather than an assumed temperature inferred from a fit to one correlator.

5 Executed checks and error scope

We evaluated (6) at $q = 1/4, 1/2$, with $\lambda = -\log q/4$, using two independent representations: the q-Hermite Jacobi matrix and the angular q-Pochhammer density and Mehler product. Each used orders 128 and 192. The 22 comparisons cover real modular times $0, .5, 1, 2, 4$, mid-strip times $.5 - i/2, 1 - i/2, 2 - i/2$, and lower-boundary times $.5 - i, 1 - i, 2 - i$. All complex values are saved.

The maximum cross-representation absolute difference is $2.34 \cdot 10^{-14}$, and the largest within-method refinement is $1.88 \cdot 10^{-12}$. The maximum cross relative difference is $2.76 \cdot 10^{-14}$; the largest phase-cancellation factor is 72.92. Neither comparison is an interval certificate. KMS boundary exchanges are sign/implementation checks, not an independent proof of modular theory. Sum rules were compared between the two representations as well.

q	$\mathcal{W}(1)$	$-\mathcal{W}''(0)$	R_{I_+J}/F_{I_+J}
1/4	$.55776147 - .30164823i$	1.09084006	1.97039949
1/2	$.40881713 - .31551359i$	1.70797502	3.28486964

Four finite-window calculations (both signs and both q) used 32/64-point interval quadrature. Their ratios exceed the analytic lower bounds 1.57692308 and 2.32096452 respectively. The analytic incompatibility in (16) does not depend on these numerical values.

A separately controlled product tail. Truncate the density products at $n = P$ and the kernel product at $n = P - 1$, as in the code. For a general Mehler parameter r, the absolute logarithmic error of the joint density is bounded by

$$\delta_P = \frac{1}{1-q}\left[\frac{6q^{P+1}}{1-q^{P+1}} + \frac{r^2 q^P}{1-r^2 q^P} + \frac{4rq^P}{1-rq^P}\right]. \tag{17}$$

This follows termwise from $|\log(1-z)| \le |z|/(1-|z|)$, including each conjugate factor. For real s, the finite-product continuous integral differs from (6) by at most $e^{\delta_P} - 1$; its positive reference weight $p(x)d\nu$ has mass one. For $s = t - i\theta$, $0 \le \theta \le 1$, the absolute weight is $p(x)^{1-\theta}p(y)^{\theta}d\nu$, whose mass is at most one by weighted arithmetic–geometric mean. The same bound holds. For the second sum rule multiply by Ω^2. We used $P = 192$. This bound excludes angular quadrature, Jacobi truncation and floating-point error.

All computations are fixed-q diagnostics. The logarithmic flow and its bounds are not uniform as $\lambda \downarrow 0$; no small-λ modular limit is asserted.

6 What this changes in the bulk construction

The constructive output is a definite preparation vector $\rho^{1/2}$ and its exact modular spectral response. It allows us to keep three objects separate in the next comparison: the trace background used in the DSSYK proposal, the prepared state used in our bilocal calculation, and the generator of candidate bulk boost time. A stationary charged restriction does not remove the need to match the preparation: its energy-filtered balance test still differs from a constant-temperature Gibbs law.

For a vacuum wedge in relativistic QFT, the modular/boost identification uses the wedge algebra and the vacuum, with the fixed 2π factor [5, Sec. 5]. Applying this statement to an arbitrary selected state and the entire microscopic algebra would change both inputs. Here $K = k(v)$ is explicitly known, but taking K to be the geometrical boost is not the dictionary.

The next constructive target is the action of a candidate null translation on the *prepared* cyclic states, with the boost still implemented by the original Hamiltonian time. The earlier translation note [8] supplies a check: two-point boost kernels alone admit arbitrary masses, while reverse-null response probes mass moments. We should calculate generator matrix elements with the bra/ket preparation kept explicit, verify the mixed boost/translation Ward identities, and only then compare the mass operator to the existing 't Hooft target. This stage supplies the state and balance data for that test; the null generator, selected four-bilocal dynamics and finite-G dictionary remain to be constructed.

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